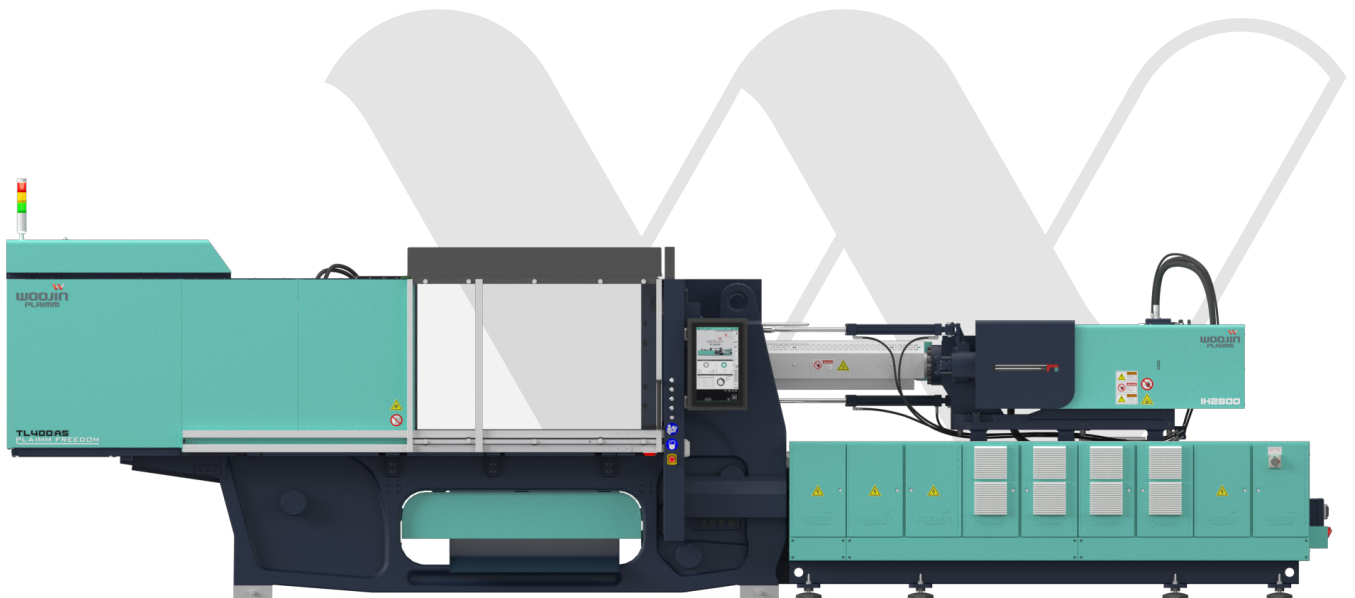


ENGLISH

TL-AS

Tie bar-Less Series




WOJIN
PLIMM

OVERVIEW

* The Red Dot Award is a globally prestigious design award, and together with Germany's iF Design Award and the USA's IDEA, it is recognized as one of the world's top three design awards

ITL-A5

Tie bar-less Series



IMM Promotion

Injection molding machine without tie-bars, jointly developed with WOJIN PLAIMM GmbH, patented by the company, and recipient of the Red Dot Award

(220~400Ton)



Hybrid Type
(Servo Motor)



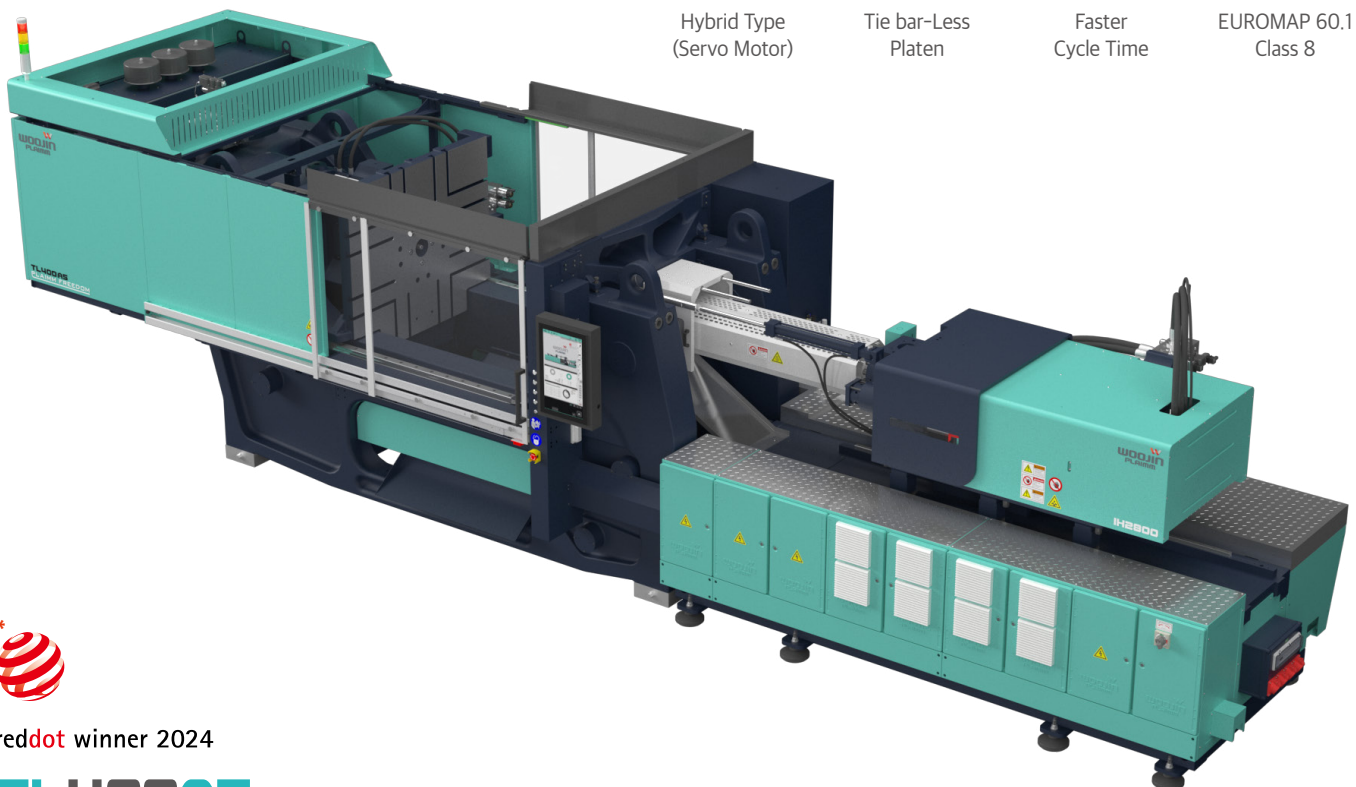
Tie bar-Less
Platen



Faster
Cycle Time



EUROMAP 60.1
Class 8



red dot winner 2024

TL400A5

※ Product images may be subject to change without prior notice due to optional specifications or product improvements.

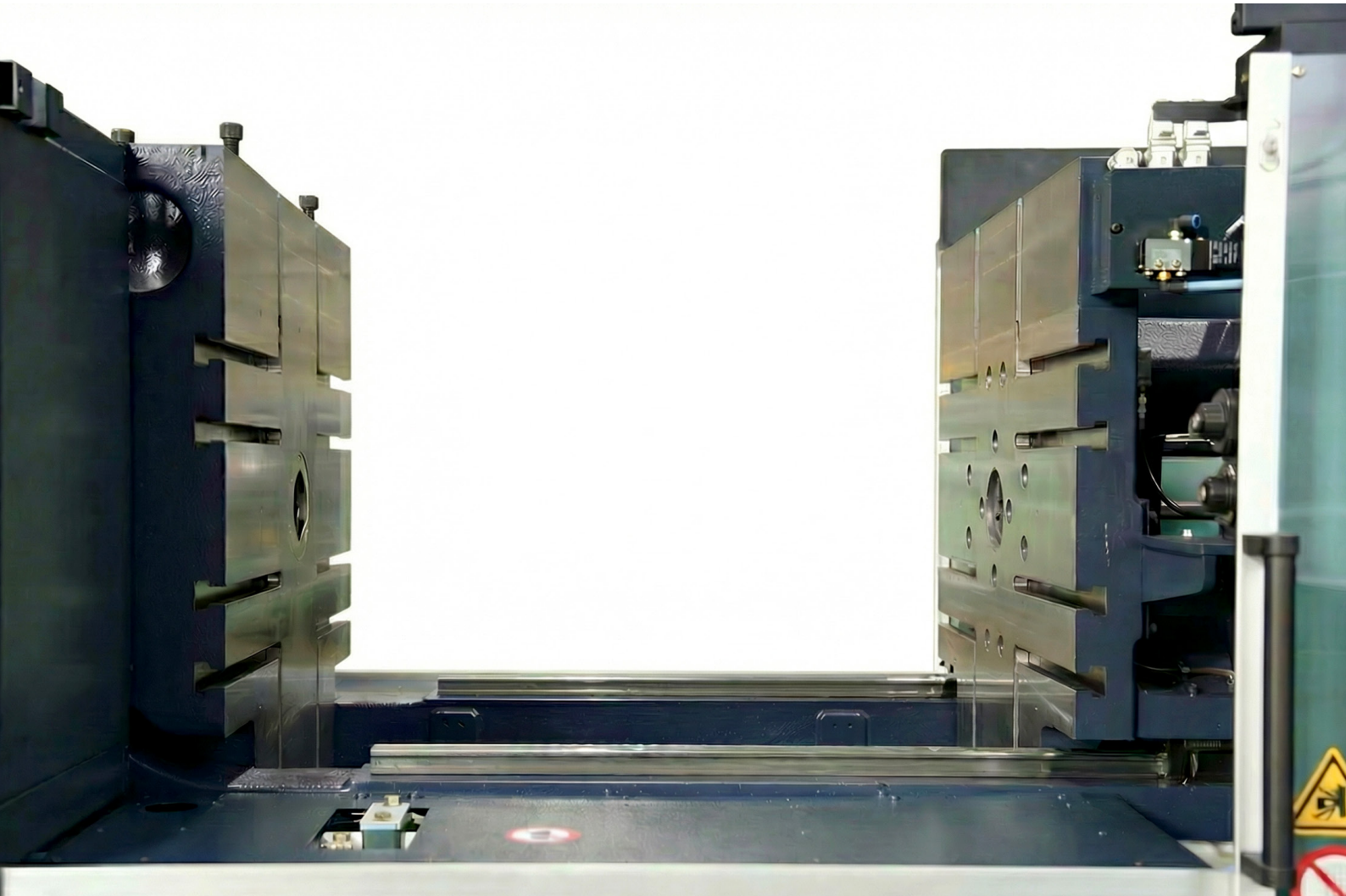
Make your dreams come true with a Tie bar-Less IMM

Co-developed with WOJIN PLAIMM GmbH (Austria)



FEATURES

Tie bar-Less

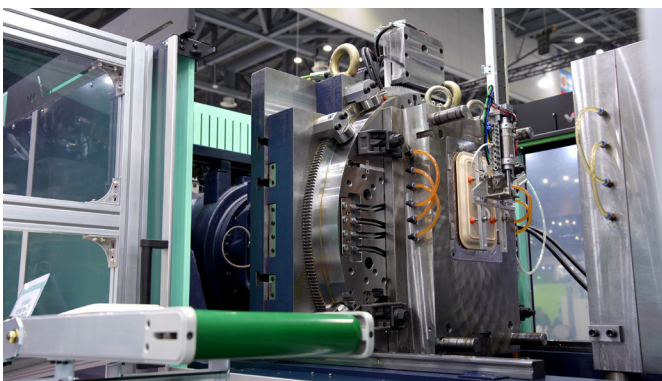


1 Wider Mold Installation Area

- | Entire platen area can be used as mold installation space
- | Approximately 30% increase in space utilization compared to tie-bar injection molding machines
- | Allows larger molds to be mounted, enabling production of larger products

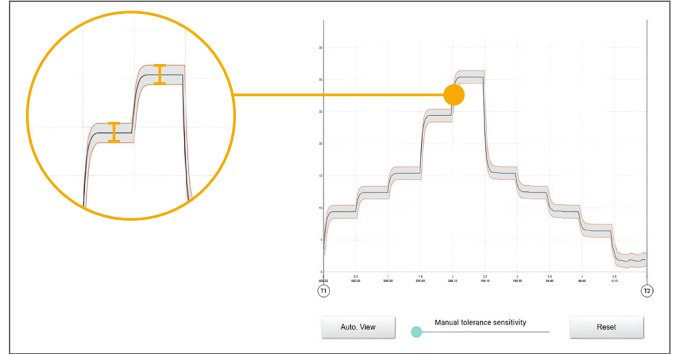
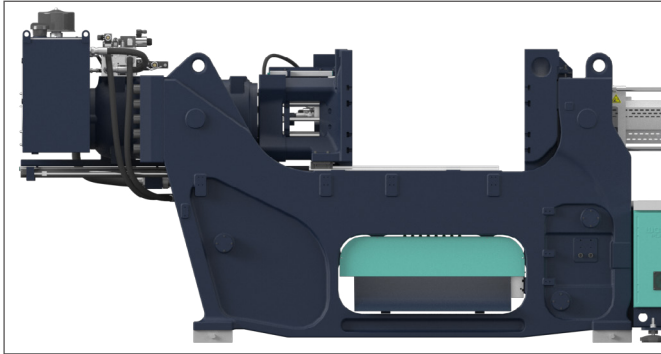
2 Shorter Cycle Time

- | Take-out robots and peripheral equipment can directly access the mold
- | Suitable for installation and operation in low-ceiling workspaces



FEATURES

MOLD PROTECTION DESIGN

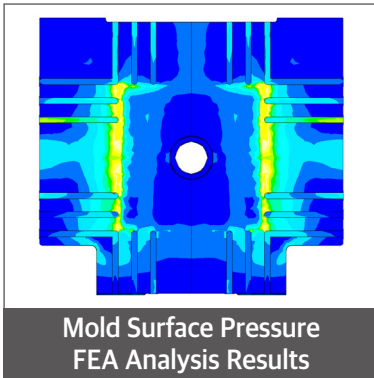


1 Mechanical Aspect

- | Centralized and distributed clamping force minimizes platen bending
- | Perfect acceleration and deceleration control possible via servo drive system
- | Equipped with high-precision sensors for precise control
- | High-response proportional valves achieve precise mold position control

2 Software Aspect (Controller Option)

- | Generates a standard graph throughout the entire injection cycle to establish a reference for good products
- | Immediately monitors foreign substances or other mold-related factors that may affect the standard graph
- | Achieves ideal “mold protection” during the mold closing phase



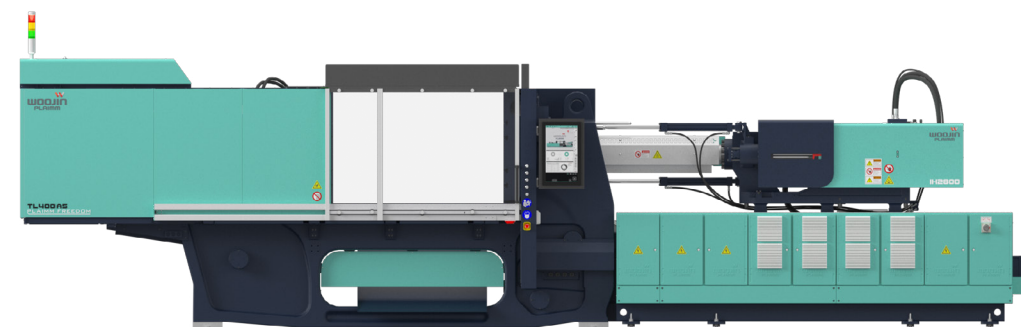
UNIFORM MOLD SURFACE PRESSURE

1 Center-Press Integrated Platen Structure

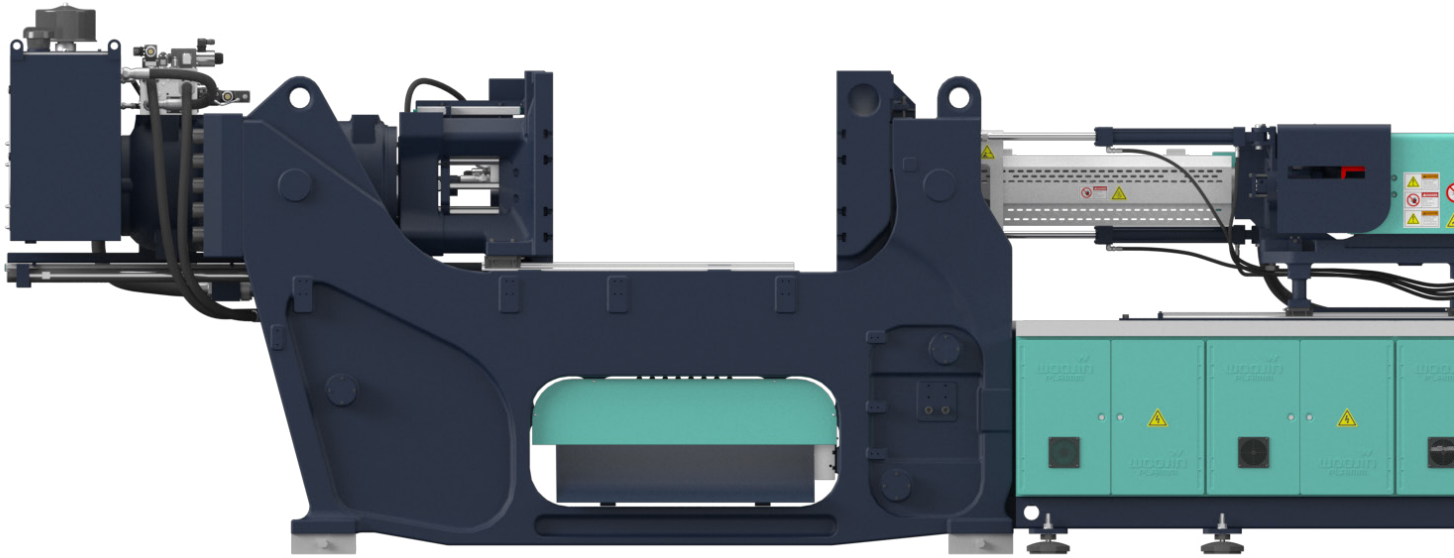
- | Designed to ensure uniform clamping force distribution across the platen
- | Evenly transmits pressure within the mold, preventing platen damage and extending mold life
- | Maintains uniform internal mold surface stress to prevent flash formation

ENERGY CONSUMPTION CLASS

- | Achieves “Class 8” energy consumption of 0.38 kWh/kg or less based on EUROMAP 60.1
- | Reduces electricity costs by more than 50% compared to conventional hydraulic systems using three-phase induction motors

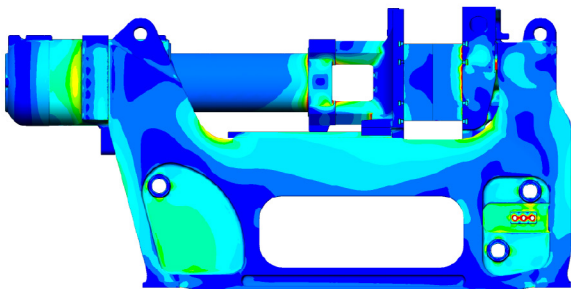


EXCELLENT PLATEN PARALLELISM



1 FEA-Optimized Design

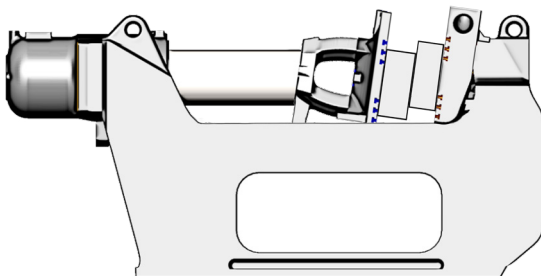
- | Tie-bar-less dedicated frame created using casting method
- | All major individual components undergo FEA



TL-A5 Clamping Unit FEA Analysis Results

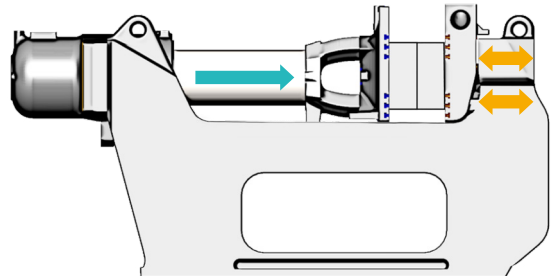
2 Two-Stage Reinforcement Structure

- | Frame structure supports the upper and middle fixed platen, while the patented “Flex Plate,” unique to WOOJIN PLAImm, supports the lower middle platen
- | Maintains ideal platen parallelism under any environment or condition (meets less than half of Euromap 09 recommended values)
- | Co-developed with WOOJIN PLAImm GmbH and patented
- | Red Dot Award 2024 Winner



[Without Patented Technology]

Maintaining platen parallelism is **critical** due to clamping force application in tie-bar-less frame design



[WOOJIN PLAImm Tie-Bar-Less Frame]

Maintains ideal platen parallelism **under any conditions** using **proprietary** patented technology

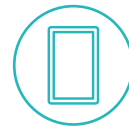
CONTROLLER

IMC 510



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FEATURES



- Equipped with B&R Controller
- 21-inch Full HD Touch Screen
- User Interface (UI/UX) Optimized for IMM



- Injection process design using Smart Sequence
- Review processes and receive feedback through PLAIMM-X

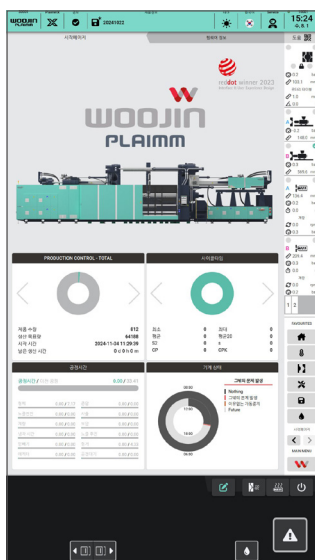


- Process management analysis and energy monitoring

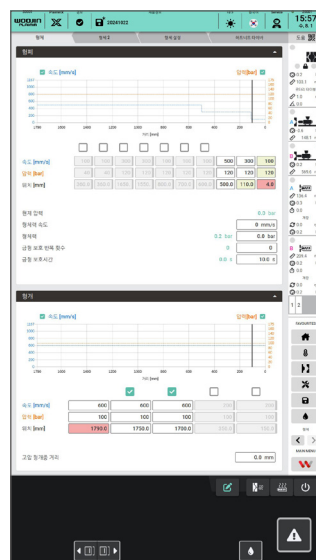


- Supports OPC-UA (EUROMAP 77)
- Data provided via SQL and CSV
- Peripheral device monitoring and remote support services available
- Check operation sequence with Overview Chain
- View clamping and injection graphs on a single page
- Frequently used functions managed through custom tabs

MAIN PAGE SCREEN CONFIGURATION



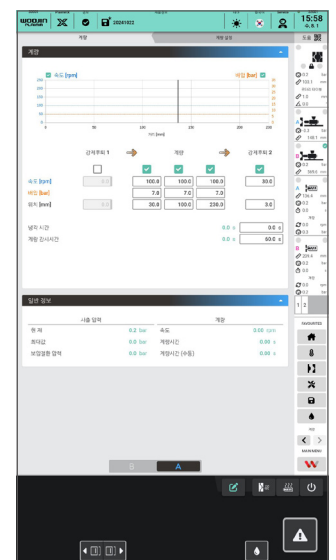
Home



Mold



Injection/Pressure Hold



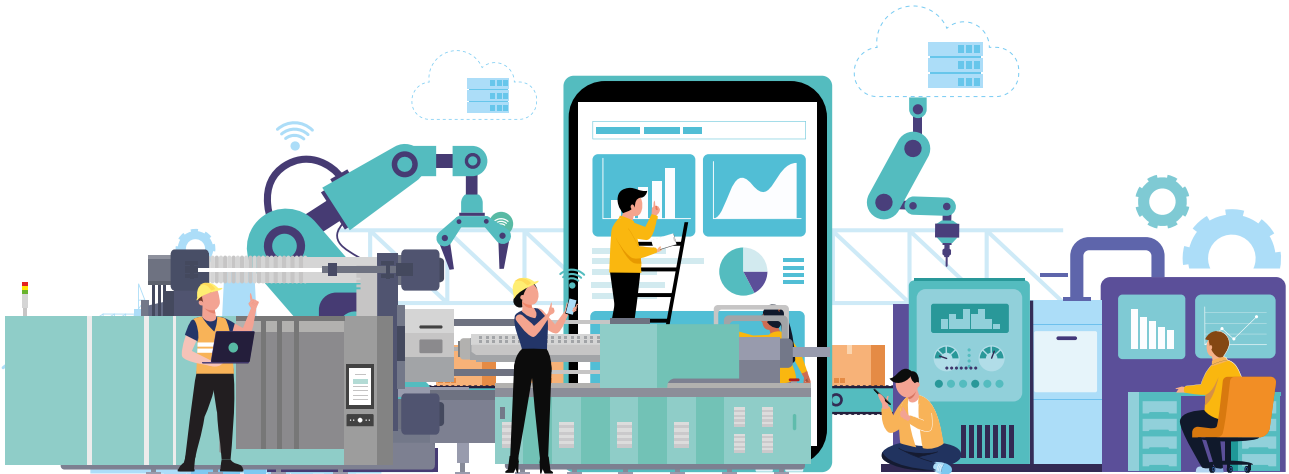
Metering

※ This controller may be changed without prior notice due to optional specifications or software updates.

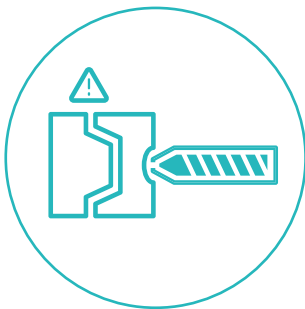
SMART INJECTION TECHNOLOGY



(OPTION)



※ Smart injection molding functions may be changed without prior notice due to software updates or product improvements.



1 Flash Monitoring

A function that detects mold flash occurring during the injection molding process to improve and stabilize product quality

- Direct clamping type: Flash measured by tie-bar position sensor
- Toggle type: Flash measured by clamping force sensor



2 Weight Control

A function that automatically adjusts holding pressure and metering process settings based on Viscosity Monitoring to minimize injection weight variations caused by changes in molten resin viscosity

Benefits

- ① Improved quality stability when molding recycled resins or eco-friendly specialty materials
- ② Reduced scrap and improved productivity through defect reduction
- ③ Supports environmental initiatives such as carbon neutrality and SDGs



3 Smart Sequencer

A function that enables flexible operation configuration in the automatic mode of the injection molding machine

Allows easy setup of special operation sequences and automatically calculates axis interlocks to ensure equipment safety

SMART INJECTION TECHNOLOGY



4 Edge Controller

- | Data integration converter function for peripheral devices via 4-channel RS485 communication
- | Connectable with various peripheral equipment such as mold temperature controllers, dryers, chillers, and hot runner controllers
- | Capable of collecting peripheral device data and transmitting it to a cloud server



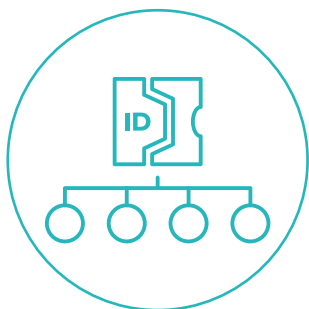
5 Energy Monitoring

- | Displays energy consumption separately for hydraulic or electric systems and cylinder heating
- | Measures and displays energy consumption by each stage of the injection molding process (Mold Closing / Injection / Holding / Metering / Cooling / Mold Opening / Ejector)
- | Helps reduce power consumption by comparing energy usage during production and downtime



6 Machine Test

- | A function that allows easy inspection and testing of the injection molding machine status
- | Enables periodic machine maintenance even for first-time users
- | Provides test functions for clamping force, injection pressure, and injection speed (additional Machine Test items planned)



7 Mold Recognition

(Coming Soon)

- | A function that enables one-touch setup of recipes for the injection molding machine, take-out robot, and peripheral equipment using mold ID information
- | Mold ID can be selected from RF-ID, QR code, or barcode



8 Data Collection Function

(Coming Soon)

- | A function that provides detailed time-series data from the injection molding machine
 - Data requiring detailed analysis
 - ① Actual and set values of injection pressure, speed, and position
 - ② clamping pressure, speed, position, and four tie-bar data
 - ③ Includes 2K (Injection A / B) data
 - Data with minimal variation
 - ① Temperature data (nozzle, barrel, hopper, hydraulic oil, etc.)
 - ② includes shot-based data for quality analysis

SPECIFICATIONS

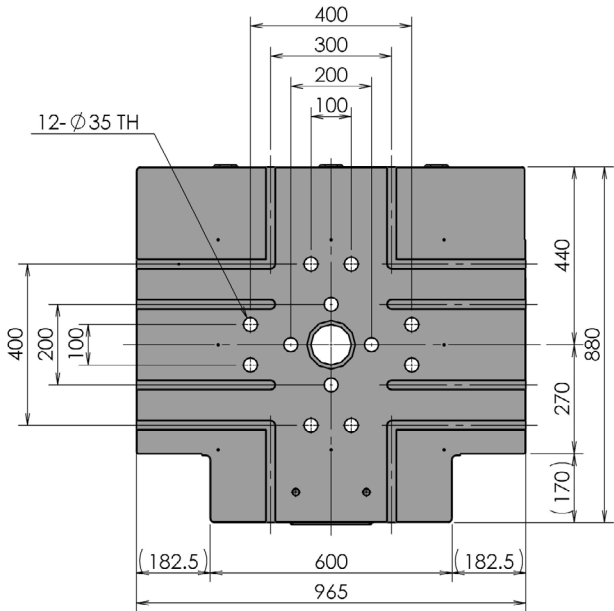
01. Theoretical injection volume: cross section of screw*screw stroke.
02. The metering distance is recommended in the range of 1 to 3 times of screw diameter.
03. Machine dimensions may vary when the injection unit moves UP or DOWN.

MODEL		TL220A5			TL300A5			TL400A5		
Injection Unit		IH1000			IH1800			IH2800		
Screw & Barrel type		O	A	B	O	A	B	O	A	B
Screw diameter	mm	45	50	55	55	60	65	65	70	80
Injection pressure	kg/cm²	2600	2258	1866	2494	2257	2008	2375	2048	1568
	Mpa	255	221	183	245	221	197	233	201	154
Theoretical injection volume	cm³	366	452	546	677	806	946	1161	1347	1759
Shot weight (PS)	g	337	416	504	624	743	871	1070	1241	1621
Injection rate	cm³/s	175	217	262	249	296	347	313	363	474
Screw stroke	mm	230			285			350		
Injection speed	mm/s	110			105			94		
Plasticizing capacity (PS)	kg/h	110	148	189	160	205	253	201	244	347
Screw rotation speed	rpm	260			220			175		
Clamping Unit										
Clamping force	ton(kN)	220(2157)			300(2941)			400(3922)		
Platen dimension (H×V)	mm	960 x 880			1120 x 1000			1250 x 1100		
Daylight	mm	800			900			1000		
Max. Daylight	mm	1100			1300			1450		
Min. mold height	mm	300			400			450		
Ejector force	ton(kN)	7.6(74.5)			9.1(89.2)			11.3(110.8)		
Ejector stroke	mm	180			200			250		
General										
Heater capacity	kW	14.6	17.1	18.7	21.0	23.8	25.7	18.4	20.6	24.1
Motor capacity	kW	30.0			55.0			55.0		
Total electric power capacity	kW	44.6	47.1	48.7	76.0	78.8	80.7	73.4	75.6	79.1
Main Oil Tank Capacity	L	370			480			525		
Clamp Oil Tank Capacity	L	225			380			440		
Total Oil Tank Capacity	L	595			860			965		
Machine weight	ton	14			19.5			25.5		
Machine dimension (L×W×H)	m	7.1 x 1.8 x 2.2			7.9 x 2.0 x 2.3			8.6 x 2.2 x 2.3		
Cooling water consumption	L/min	40			65			65		

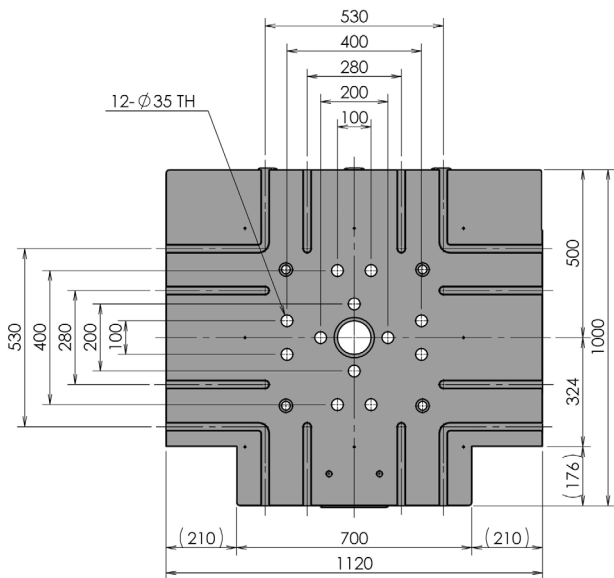
PLATEN DIMENSION

Unit: mm

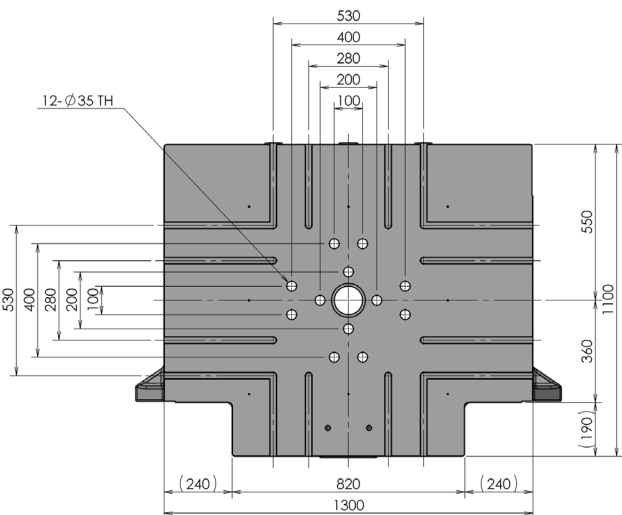
TL220A5



TL300A5

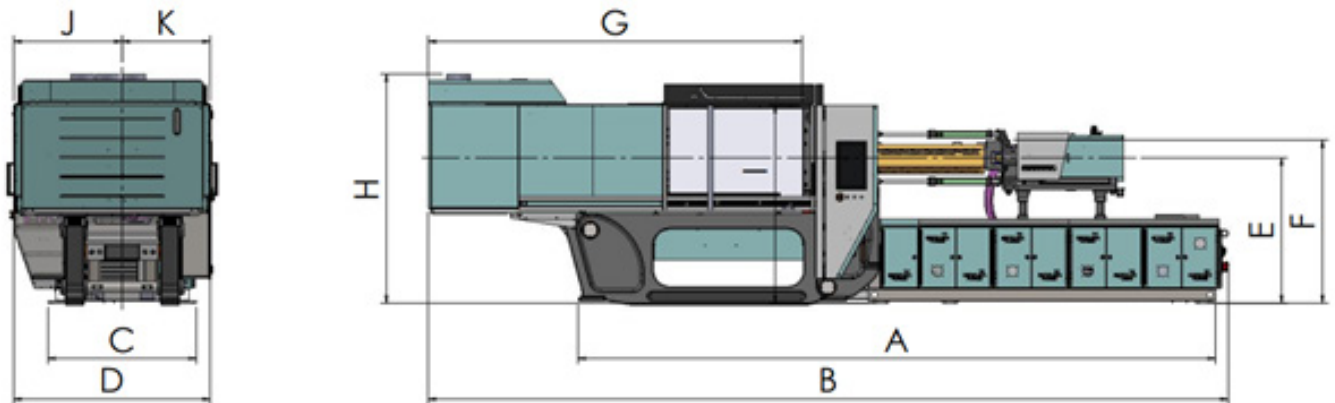


TL400A5



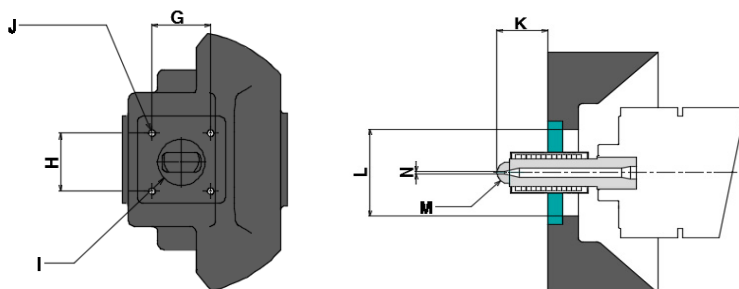
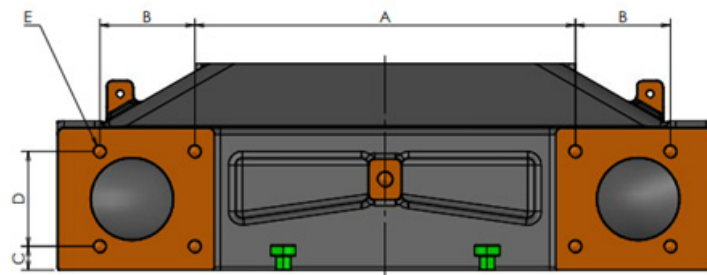
※ Product images may be subject to change without prior notice due to optional specifications or product improvements.

MACHINE DIMENSION



Machine Dimension

Unit: mm	A	B	C	D	E	F	G	H	I	J	K
TL220A5	5710	7079	1450	1771	1349	1529	3271	2121	1794	965	806
TL300A5	6230	7905	1450	1923	1425	1600	3658	2222	1940	1060	863
TL400A5	6900	8560	1450	2166	1440	1630	4113	2250	1995	1210	956

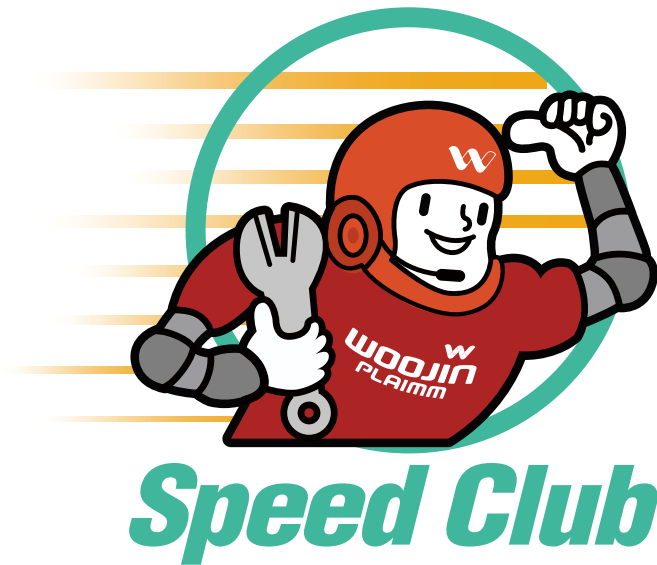


Unit: mm	Robot installation position dimension					Hopper installation position dimension			Nozzle dimension				
	A	B	C	D	F	G	H	ØI	J	K	ØL	M	ØN
TL220A5	560	140	35	140	8-M20 TAP DP40	120	120	100	4-M12 TAP	50	100	R14	3.5
TL300A5	700	140	35	140	8-M20 TAP DP40	127	127	115	4-M12 TAP	50	100	R14	3.5
TL400A5	840	140	45	140	8-M24 TAP DP48	127	127	115	4-M12 TAP	120	100	R19	4

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OPTION LIST

Injection Unit	Clamping Unit	General
Standard		
01. Single Flight Screw 02. Injection valve gate circuit (AC 1 + DC 1) 03. Back-Pressure Closed-loop system 04. PID Heating Control 05. Weekly Heating Timer 06. Cold screw start protection mode 07. Temperature display & Alarm in abnormal Temp. 08. Auto Purging 09. Injection Speed & Pressure step (10 step) 10. Holding Speed & Pressure step (5 step) 11. Charging Speed & Pressure step (3 step) 12. Back Pressure control step (3 step) 13. Injection Pressure Graph Display 14. Injection Speed Graph Display 15. Screw RPM Display 16. Cushion Display & Alarm 17. Charging time count & alarm 18. Screw & Barrel (Anti Wear)	01. Mold thickness adjusting break unit 02. Hydraulic Core puller (Moving platen side, 1 stage) 03. Air blow-off unit (Fixed side 1 + Moving side 1) 04. Safety device(for electric & hydraulic) 05. Automatic Mold thickness adjust mode 06. Mold-Open Speed & Pressure step (4 step) 07. Mold-Close Speed & Pressure step (5 step) 08. Ejector Speed & Pressure step (2 step)	01. Standard Maintenance tools 02. Standard spare part 03. Leveling pad 04. Cooling water distributor 05. Automatic oil lubrication (Toggle) 06. Robot interface (Standard) 07. 3 Phase electric outlet (2 ea) 08. Single Phase electric outlet (1 ea) 09. Hopper throat temperature control device 10. Hydraulic oil purification device 11. Hydraulic oil temperature control device 12. Hydraulic oil level alarm 13. Hydraulic oil temperature check & alarm 14. Hydraulic oil heating mode 15. 3 color alarm light 16. Shot data saving by external way 17. Production data statistics 18. Alarming & History save 19. Log history save 20. I/O circuit display 21. Shot data save (Internal 1,000 / External device) 22. Steel tray for resin leakage
Option		
01. Heater Disconnection check device 02. Hopper Slide (L/M) 03. Screw & Barrel (Nitrided barrel) 04. Screw & Barrel (Anti Wear & Corrosive) 05. Fast Injection Circuit (ACC) 06. Valve Gate Circuit & Connector (Interior type) 07. Hydraulic Valve Gate Block (Interior type) 08. Pneumatic Valve Gate Block (Interior type) 09. Hydraulic Valve Gate Device (External device) 10. Charging on Fly (AC Motor) 11. Shut-off Nozzle 12. Customized Design Screw (SB, Mixing, Coating)	01. Rotating Core Circuit 02. Core & Ejector on Fly 03. Daylight Extension 04. Automatic safety Door open/close 05. Core Pressure release Circuit (Automatic) 06. Product Chute 07. Hydraulic Core Check Valve 08. Hydraulic Core Interlock Connector (EM13, WJ Standard) 09. Hydraulic Core Puller (Fixed, 1~4 Stages) 10. Hydraulic Core Puller (Moving 2~4 Stages) 11. Mold ring on Moving-Platen 12. Ejector Check Valve 13. Ejector Interlock Connector (WJ Standard, EM13) 14. Ejector Forward/Backward External switch 15. Mold Insulation Platen 16. Pneumatic Core Puller (1-3 Stages)	01. Lubricating oil Recycling device 02. Product drop check device 03. Product quality sorting device (Below 280ton) 04. Hydraulic Auto-Clamp unit 05. Heater Insulation Band 06. Automatic Grease Lubrication (Clamping unit) 07. Robot Interface (EM12, EM67, EM67.1, SPI) 08. CMS (Central Monitoring System) 09. AVR (Automatic Voltage Regulator) on Electric Panel 10. UPS (Uninterruptible Power Supply) on Electric Panel 11. Dosing unit Interface (for Masterbatch) 12. Gas Injection Interface 13. Steam Injection Interface 14. External Temperature Display (F/P) 15. Interior type Hot Runner Controller (EM13, WJ Standard)



- 1 24-Hour On-Site Support
- 2 Dedicated B/S Team
- 3 Mobile App Support
- 4 Extended Parts Warranty

1 Expert Technical Staff

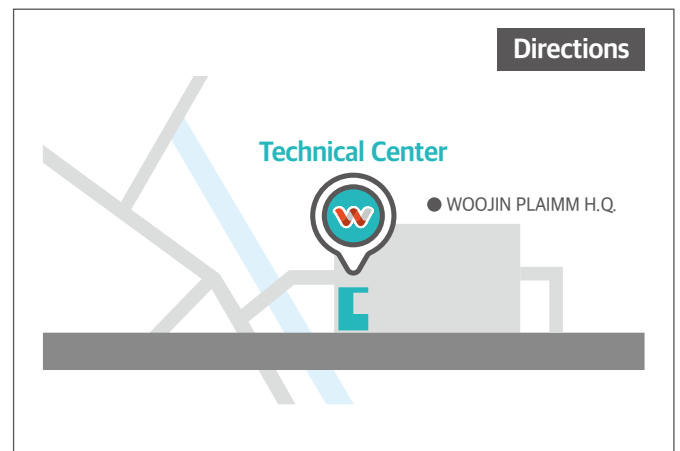
| Over 10 Years of Experience in Injection Molding
(Optimization / Foaming / Trial Injection)

2 Technical Research Team

| Control technology and mechanical design personnel
available for ongoing support

3 Auxiliary Facility Support

| Deployment of Skilled Technicians and Consulting Staff



WOOJIN PLAIMM Technical Center **delivers the best results**
to our customers through **strategic consulting**
and **technical knowledge sharing.**



GLOBAL NETWORK



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HOMEPAGE



YOUTUBE



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Korea No.1 IMM Manufacturer

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